

Hyperuniformity in Chemotactic Active Matter

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1 Models

1.1 Particle-based model

$$\frac{d}{dt}\mathbf{r}_i(t) = \alpha \nabla c(\mathbf{r}_i(t), t) + \sqrt{2D}\boldsymbol{\xi}_i(t) , \quad (1a)$$

$$\frac{\partial}{\partial t}c(\mathbf{r}, t) = D_c \nabla^2 c(\mathbf{r}, t) + \mu c(\mathbf{r}, t) + k \sum_{j=1}^N \delta(\mathbf{r} - \mathbf{r}_j(t)) , \quad (1b)$$

for $i = 1, 2, \dots, N$. Here, $\mathbf{r}_i$ is the position of the i -th particle, c is the concentration field of the chemoattractant, α is the chemotactic sensitivity. If $\alpha > 0$, the particles are attracted to higher concentrations of the chemoattractant (chemoattraction/positive chemotaxis), while if $\alpha < 0$, the particles are repelled by higher concentrations of the chemoattractant (chemorepulsion/negative chemotaxis). D is the diffusion coefficient of the particles, $\boldsymbol{\xi}_i$ is an independent Gaussian white noise with zero mean and unit variance for each particle, D_c is the diffusion coefficient of the chemoattractant, μ is the decay ($\mu < 0$) or growth ($\mu > 0$) rate of the chemoattractant, and k is the production ($k > 0$) or consumption rate ($k < 0$) of the chemoattractant by the particles.

For simplicity, we assume that particles are point-like and move in a two-dimensional $L \times L$ square domain with periodic boundary conditions. The initial positions of the particles can be randomly distributed, and the initial concentration field can be set to a small random perturbation around a uniform background concentration c_0 .

To reduce the parameter space, we can non-dimensionalize the equations by introducing $x_0 = L$, $t_0 = 1/k$ and c_0 as characteristic scales of length, time and concentration. The resulting dimensionless parameters are $\tilde{\alpha} = \frac{\alpha c_0}{L^2 k}$, $\tilde{D} = \frac{D}{L^2 k}$, $\tilde{D}_c = \frac{D_c}{L^2 k}$, and $\tilde{\mu} = \frac{\mu}{k}$. We can omit the tildes and work with the dimensionless equations:

$$\frac{d}{dt}\mathbf{r}_i(t) = \alpha \nabla c(\mathbf{r}_i(t), t) + \sqrt{2\tilde{D}}\boldsymbol{\xi}_i(t) , \quad (2a)$$

$$\frac{\partial}{\partial t}c(\mathbf{r}, t) = \tilde{D}_c \nabla^2 c(\mathbf{r}, t) + \tilde{\mu} c(\mathbf{r}, t) + \sum_{j=1}^N \delta(\mathbf{r} - \mathbf{r}_j(t)) . \quad (2b)$$

Note that the above equations mathematically implies the assumption of $k > 0$. If $k < 0$, the non-dimensionalization would be different, and the Eq. (2b) would have a negative sign in front of the summation term:

$$\frac{\partial}{\partial t} c(\mathbf{r}, t) = D_c \nabla^2 c(\mathbf{r}, t) + \mu c(\mathbf{r}, t) - \sum_{j=1}^N \delta(\mathbf{r} - \mathbf{r}_j(t)) . \quad (3)$$

In the following, we focus on the case $k < 0$, corresponding to chemical consumption. For $\alpha > 0$, consumption lowers the chemical concentration around particles and thereby generates an effective repulsion between them, which is more relevant to the suppression of density fluctuations.

As shown in the snapshots, the case of production ($k > 0$) leads to aggregation of particles and formation of dense clusters, while the case of consumption ($k < 0$) leads to dispersion of particles and formation of more uniform distributions. The latter case is more likely to exhibit hyperuniformity.

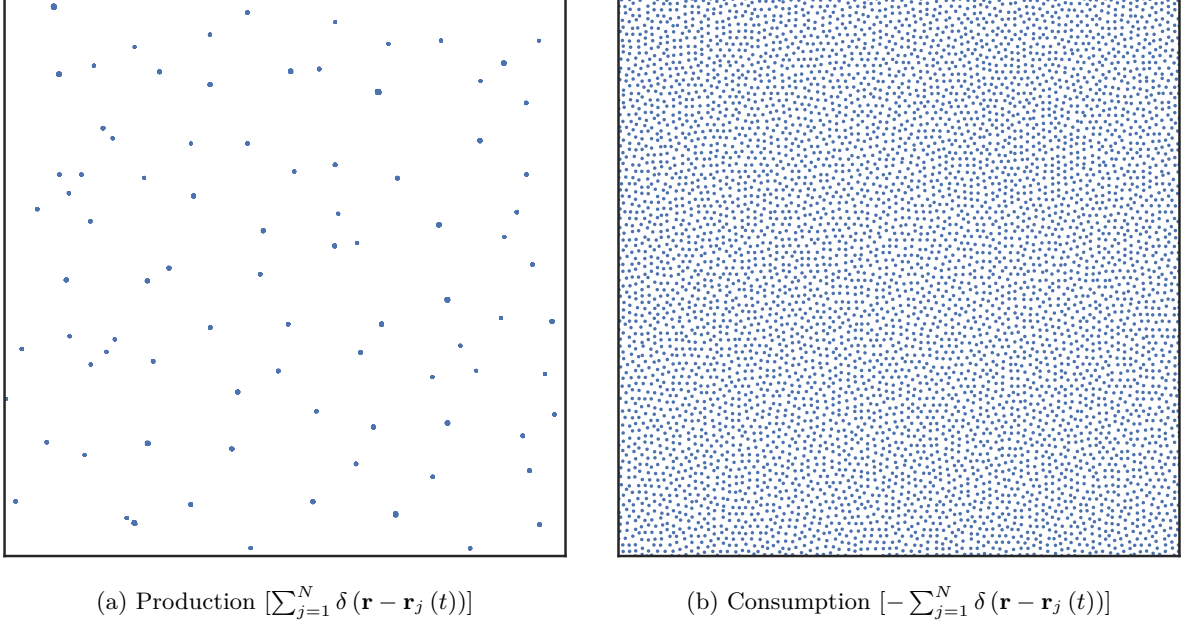


Figure 1: Snapshots of particle configurations for (a) production and (b) consumption of chemoattractant. The parameters are $\alpha = 5$, $D = 0.0001$, $D_c = 5$, $\mu = -0.0001$, and $N = 6400$ particles in an $L = 20$ domain. The positions and velocity direction of the particles are shown as arrows.

1.2 Continuum model

It is convenient to describe collective behavior using a particle density field $\rho(\mathbf{r}, t) = \sum_{i=1}^N \delta(\mathbf{r} - \mathbf{r}_i(t))$ (Note that the $\rho(\mathbf{r}, t)$ here is particle number density, not probability density). By applying the Dean's method [1], we can derive the following stochastic partial differential equations for ρ and c :

$$\frac{\partial}{\partial t} \rho(\mathbf{r}, t) = -\alpha \nabla \cdot (\rho \nabla c) + D \nabla^2 \rho - \nabla \cdot [\sqrt{2D\rho} \mathbf{\Lambda}] , \quad (4a)$$

$$\frac{\partial}{\partial t} c(\mathbf{r}, t) = D_c \nabla^2 c + \mu c - \rho , \quad (4b)$$

where $\mathbf{\Lambda}(\mathbf{r}, t)$ is a Gaussian white noise with zero mean and unit variance, satisfying

$$\langle \Lambda_a(\mathbf{r}, t) \Lambda_b(\mathbf{r}', t') \rangle = \delta_{ab} \delta(\mathbf{r} - \mathbf{r}') \delta(t - t') . \quad (5)$$

Under the mean-field approximation, $\langle \mathbf{\Lambda} \rangle = 0$ leads to $\langle -\nabla \cdot [\sqrt{2D\rho} \mathbf{\Lambda}] \rangle \approx 0$, and we can neglect

the noise term. This gives us the classical Keller-Segel model [2, 3]:

$$\frac{\partial}{\partial t} \rho(\mathbf{r}, t) = -\alpha \nabla \cdot (\rho \nabla c) + D \nabla^2 \rho, \quad (6a)$$

$$\frac{\partial}{\partial t} c(\mathbf{r}, t) = D_c \nabla^2 c + \mu c - \rho. \quad (6b)$$

One can perform linear stability analysis on the Keller-Segel model to determine the conditions for pattern formation. However, the mean-field approximation neglects fluctuations, which can be crucial for understanding the emergence of hyperuniformity. Therefore, we need to consider the full stochastic equations with noise to capture the effects of fluctuations on the system's behavior.

One obvious solution of Eqs. (6) is the homogeneous state $\rho(\mathbf{r}, t) = \rho_0$ and $c(\mathbf{r}, t) = c_0$, where $c_0 = \rho_0/\mu$, which can be regarded as solution to Eqs. (4) in the mean-field sense. To analyze the stability of this homogeneous state, we can introduce small perturbations $\delta\rho(\mathbf{r}, t)$ and $\delta c(\mathbf{r}, t)$ around the homogeneous solution:

$$\rho(\mathbf{r}, t) = \rho_0 + \delta\rho(\mathbf{r}, t), \quad c(\mathbf{r}, t) = c_0 + \delta c(\mathbf{r}, t). \quad (7)$$

Introducing these perturbations into Eqs. (4) gives us the following linearized equations:

$$\frac{\partial}{\partial t} \delta\rho = -\alpha \rho_0 \nabla^2 \delta c + D \nabla^2 \delta\rho - \nabla \cdot [\sqrt{2D\rho_0} \mathbf{\Lambda}] , \quad (8a)$$

$$\frac{\partial}{\partial t} \delta c = D_c \nabla^2 \delta c + \mu \delta c - \delta\rho. \quad (8b)$$

By performing Fourier transform on the linearized equations, we can analyze the growth rates of different modes and determine the conditions for instability. The presence of noise can lead to fluctuations that may drive the system towards hyperuniform states, even in regimes where the mean-field analysis predicts stability. Therefore, it is essential to consider the full stochastic dynamics to understand the emergence of hyperuniformity in chemotactic active matter.

Let $V = L^d$ denote the system volume. Under periodic boundary conditions, the allowed wavevectors are $\mathbf{k} = 2\pi\mathbf{n}/L$. We use the symmetric Fourier convention

$$\delta\rho(\mathbf{r}, t) = \frac{1}{\sqrt{V}} \sum_{\mathbf{k}} \delta\rho_{\mathbf{k}}(t) e^{i\mathbf{k} \cdot \mathbf{r}}, \quad \delta\rho_{\mathbf{k}}(t) = \frac{1}{\sqrt{V}} \int_V d\mathbf{r} e^{-i\mathbf{k} \cdot \mathbf{r}} \delta\rho(\mathbf{r}, t), \quad (9a)$$

$$\delta c(\mathbf{r}, t) = \frac{1}{\sqrt{V}} \sum_{\mathbf{k}} \delta c_{\mathbf{k}}(t) e^{i\mathbf{k} \cdot \mathbf{r}}, \quad \delta c_{\mathbf{k}}(t) = \frac{1}{\sqrt{V}} \int_V d\mathbf{r} e^{-i\mathbf{k} \cdot \mathbf{r}} \delta c(\mathbf{r}, t). \quad (9b)$$

The factor $1/\sqrt{V}$ is chosen so that the Fourier-space density covariance has the same units as the mean number density. With this convention, the corresponding Fourier component of the white-noise field is

$$\mathbf{\Lambda}_{\mathbf{k}}(t) = \frac{1}{\sqrt{V}} \int_V d\mathbf{r} e^{-i\mathbf{k} \cdot \mathbf{r}} \mathbf{\Lambda}(\mathbf{r}, t), \quad (10)$$

and

$$\nabla \cdot [\sqrt{2D\rho_0} \mathbf{\Lambda}(\mathbf{r}, t)] = \frac{1}{\sqrt{V}} \sum_{\mathbf{k}} i\mathbf{k} \cdot \sqrt{2D\rho_0} \mathbf{\Lambda}_{\mathbf{k}}(t) e^{i\mathbf{k} \cdot \mathbf{r}}. \quad (11)$$

After the Fourier transform, the linearized equations become:

$$\frac{\partial}{\partial t} \delta\rho_{\mathbf{k}} = -Dk^2 \delta\rho_{\mathbf{k}} + \alpha \rho_0 k^2 \delta c_{\mathbf{k}} + \eta_{\mathbf{k}}, \quad (12a)$$

$$\frac{\partial}{\partial t} \delta c_{\mathbf{k}} = -(D_c k^2 - \mu) \delta c_{\mathbf{k}} - \delta\rho_{\mathbf{k}}, \quad (12b)$$

where $\eta_{\mathbf{k}} = -i\mathbf{k} \cdot \sqrt{2D\rho_0} \mathbf{\Lambda}_{\mathbf{k}}$. The noise has zero mean and covariance

$$\langle \eta_{\mathbf{k}}(t) \eta_{\mathbf{k}'}(t') \rangle = 2D\rho_0 k^2 \delta_{\mathbf{k}, -\mathbf{k}'} \delta(t - t'). \quad (13)$$

Thus, for the conjugate pair $\mathbf{k}' = -\mathbf{k}$, the noise strength entering the covariance of a single mode is $2D\rho_0 k^2$. In a continuous Fourier convention, the Kronecker delta is replaced by the corresponding Dirac delta in wavevector space. Eqs. (12) can be rewritten in matrix form as

$$\partial_t \begin{bmatrix} \delta\rho_{\mathbf{k}} \\ \delta c_{\mathbf{k}} \end{bmatrix} = A(k) \begin{bmatrix} \delta\rho_{\mathbf{k}} \\ \delta c_{\mathbf{k}} \end{bmatrix} + \begin{bmatrix} \eta_{\mathbf{k}} \\ 0 \end{bmatrix}, \quad (14)$$

where

$$A(k) = \begin{bmatrix} -Dk^2 & \alpha\rho_0 k^2 \\ -1 & \mu - D_c k^2 \end{bmatrix}. \quad (15)$$

Let s be the linear growth rate, which is the eigenvalue of $A(k)$. The characteristic equation is given by

$$s^2 - (\text{Tr}A)s + \det A = 0. \quad (16)$$

where $\text{Tr}A = -Dk^2 + \mu - D_c k^2$ and $\det A = Dk^2(D_c k^2 - \mu) + \alpha\rho_0 k^2$. The homogeneous state is linearly unstable if there exists a wavevector k such that the real part of s is positive. This leads to the stability condition:

$$\text{Tr}A < 0, \quad \det A > 0, \quad (17)$$

which leads to the following inequalities:

$$\begin{cases} \mu < (D + D_c)k^2, \\ \mu < D_c k^2 + \frac{\alpha\rho_0}{D}. \end{cases} \quad (18)$$

Therefore, the homogeneous state is linearly stable if $\mu < \min\{(D + D_c)k^2, D_c k^2 + \frac{\alpha\rho_0}{D}\}$ for all k . In particular, for small k , the stability condition reduces to $\mu < 0 < \frac{\alpha\rho_0}{D}$, which means that the homogeneous state is stable if the chemoattractant decays and the chemotactic sensitivity is positive

$$\mu < 0, \quad \alpha > 0. \quad (19)$$

However, even in the stable regime, fluctuations can still play a significant role in shaping the spatial distribution of particles, which is crucial for understanding the emergence of hyperuniformity.

1.2.1 Density spectrum and normalized structure factor

Before calculating the fluctuation spectrum, it is important to distinguish two closely related quantities. With the Fourier convention in Eq. (9), define the density covariance spectrum

$$C_{\rho\rho}(k) \equiv \langle \delta\rho_{\mathbf{k}} \delta\rho_{-\mathbf{k}} \rangle, \quad (20)$$

and the conventional dimensionless structure factor

$$S(k) \equiv \frac{C_{\rho\rho}(k)}{\rho_0}. \quad (21)$$

For a microscopic point configuration and a nonzero allowed wavevector, this definition is equivalent to

$$S(k) = \frac{1}{N} \left\langle \left| \sum_{j=1}^N e^{-i\mathbf{k} \cdot \mathbf{r}_j} \right|^2 \right\rangle. \quad (22)$$

Consequently, a homogeneous Poisson point process has

$$C_{\rho\rho}^{\text{P}}(k) = \rho_0, \quad S^{\text{P}}(k) = 1, \quad \mathbf{k} \neq \mathbf{0}. \quad (23)$$

The strictly zero mode is excluded: in a system with fixed particle number, it is constrained by number conservation, whereas hyperuniformity concerns the limit $k \rightarrow 0$ through nonzero wavevectors.

In the limit in which the chemical field relaxes sufficiently rapidly compared with the density mode, we may set $\partial_t \delta c_{\mathbf{k}} \approx 0$ in Eq. (12), obtaining

$$\delta c_{\mathbf{k}} \approx -\frac{\delta \rho_{\mathbf{k}}}{D_c k^2 - \mu}. \quad (24)$$

Substitution into the density equation gives

$$\partial_t \delta \rho_{\mathbf{k}} = -k^2 \left[D + \frac{\alpha \rho_0}{D_c k^2 - \mu} \right] \delta \rho_{\mathbf{k}} + \eta_{\mathbf{k}}. \quad (25)$$

Defining

$$D_{\text{eff}}(k) = D + \frac{\alpha \rho_0}{D_c k^2 - \mu}, \quad (26)$$

we obtain an Ornstein–Uhlenbeck process with relaxation rate $k^2 D_{\text{eff}}(k)$. Using Appendix 2.2, its steady-state density covariance is

$$C_{\rho\rho}^{\text{qs}}(k) = \frac{2D\rho_0 k^2}{2k^2 D_{\text{eff}}(k)} = \frac{D\rho_0}{D + \frac{\alpha \rho_0}{D_c k^2 - \mu}}, \quad (27)$$

and therefore the normalized structure factor is

$$S_{\text{qs}}(k) = \frac{C_{\rho\rho}^{\text{qs}}(k)}{\rho_0} = \frac{D}{D + \frac{\alpha \rho_0}{D_c k^2 - \mu}}. \quad (28)$$

The numerator of the covariance calculation represents the strength $2D\rho_0 k^2$ of the conserved Dean noise, while the denominator contains twice the relaxation rate. The normalization in Eq. (21) removes the Poisson density scale. In particular, setting $\alpha = 0$ gives $S(k) = 1$, as required for independent Brownian particles.

For fixed $\mu < 0$, the long-wavelength limit of Eq. (28) is

$$S_0 \equiv \lim_{k \rightarrow 0} S_{\text{qs}}(k) = \frac{D}{D - \frac{\alpha \rho_0}{\mu}} = \frac{D|\mu|}{D|\mu| + \alpha \rho_0}. \quad (29)$$

Hence, in the stable regime $\mu < 0$ and $\alpha > 0$,

$$0 < S_0 < 1. \quad (30)$$

The large-scale fluctuations are suppressed relative to a Poisson process, but the nonzero plateau means that the state is not strictly hyperuniform.

Without the quasi-steady approximation, the full two-component Ornstein–Uhlenbeck system in Eq. (14) can be solved using the Lyapunov equation. As derived in Appendix 2.3, the normalized structure factor is

$$S_{\text{full}}(k) = D \frac{\alpha \rho_0 k^2 + (D_c k^2 - \mu) [(D + D_c)k^2 - \mu]}{[(D + D_c)k^2 - \mu] [D(D_c k^2 - \mu) + \alpha \rho_0]}. \quad (31)$$

This expression satisfies the consistency checks $S_{\text{full}}(k) = 1$ when $\alpha = 0$ and $S_{\text{full}}(k) \rightarrow 1$ as $k \rightarrow \infty$. For any fixed $\mu < 0$, its $k \rightarrow 0$ limit is exactly Eq. (29). The point $\mu = 0$, however, is singular and must be treated separately because the chemical relaxation rate $D_c k^2 - \mu$ then also vanishes as $k \rightarrow 0$.

1.2.2 Hyperuniformity and the order of limits

A statistically homogeneous point pattern is hyperuniform when its conventional structure factor satisfies

$$\lim_{k \rightarrow 0} S(k) = 0. \quad (32)$$

For every fixed $\mu < 0$ and finite $D > 0$, Eq. (29) gives a positive long-wavelength plateau. Thus the present linear theory predicts suppressed, but not strictly hyperuniform, density fluctuations in that parameter regime.

Two singular limits require special care. First, $D \rightarrow 0$ removes both ordinary particle diffusion and the Dean-noise amplitude. The resulting disappearance of steady-state fluctuations is therefore a trivial zero-noise limit rather than a nontrivial fluctuation-suppression mechanism.

Second, if one imposes a quasi-static chemical field and then sets $\mu = 0$, Eq. (28) becomes

$$S_{\text{qs}}(k) = \frac{DD_c k^2}{DD_c k^2 + \alpha \rho_0} \underset{k \rightarrow 0}{\sim} \frac{DD_c}{\alpha \rho_0} k^2. \quad (33)$$

Within this instantaneous, elliptic-field approximation, the model displays class-I hyperuniform scaling. However, this conclusion cannot be transferred directly to the full dynamical chemical field. At $\mu = 0$, the chemical relaxation rate is $D_c k^2$, which vanishes at long wavelength; therefore the chemical field is not a fast variable in the limit $k \rightarrow 0$, and the adiabatic elimination used above ceases to be uniformly valid.

Setting $\mu = 0$ directly in the full result, Eq. (31), gives

$$S_{\text{full}}(k; \mu = 0) = \frac{D}{D + D_c} \frac{\alpha \rho_0 + D_c(D + D_c)k^2}{\alpha \rho_0 + DD_c k^2}, \quad (34)$$

so that

$$\lim_{k \rightarrow 0} S_{\text{full}}(k; \mu = 0) = \frac{D}{D + D_c} > 0. \quad (35)$$

The limits $k \rightarrow 0$ and $\mu \rightarrow 0^-$ therefore do not commute:

$$\lim_{\mu \rightarrow 0^-} \lim_{k \rightarrow 0} S_{\text{full}}(k) = 0, \quad \lim_{k \rightarrow 0} \lim_{\mu \rightarrow 0^-} S_{\text{full}}(k) = \frac{D}{D + D_c}. \quad (36)$$

For small but nonzero $|\mu|$, a natural crossover wave number is

$$k_\mu \sim \sqrt{\frac{|\mu|}{D_c}}. \quad (37)$$

At $k \ll k_\mu$, the structure factor approaches the finite plateau in Eq. (29). Over a finite intermediate range above this crossover, the spectrum may show an apparent k^2 -like suppression if the chemical field is sufficiently fast, but this should be described as finite-scale or “hyperuniform-like” behavior rather than strict asymptotic hyperuniformity.

2 Appendix

2.1 Derivation of Dean’s equation

The continuum equation for the particle density can be derived from the microscopic Langevin dynamics by applying Itô’s formula. We start from the microscopic density

$$\rho(\mathbf{r}, t) = \sum_{i=1}^N \delta(\mathbf{r} - \mathbf{r}_i(t)). \quad (38)$$

Taking the time derivative and using the chain rule gives the exact continuity equation

$$\partial_t \rho(\mathbf{r}, t) = -\nabla \cdot \mathbf{J}(\mathbf{r}, t), \quad (39)$$

where the microscopic current is

$$\mathbf{J}(\mathbf{r}, t) = \sum_{i=1}^N \dot{\mathbf{r}}_i(t) \delta(\mathbf{r} - \mathbf{r}_i(t)). \quad (40)$$

Substituting the particle Langevin equation into the current yields two contributions. The chemotactic drift gives

$$\mathbf{J}_{\text{chemo}}(\mathbf{r}, t) = \alpha \sum_{i=1}^N \nabla c(\mathbf{r}_i, t) \delta(\mathbf{r} - \mathbf{r}_i). \quad (41)$$

Under the local-field approximation $\nabla c(\mathbf{r}_i, t) \approx \nabla c(\mathbf{r}, t)$, this becomes

$$\mathbf{J}_{\text{chemo}}(\mathbf{r}, t) \approx \alpha \rho(\mathbf{r}, t) \nabla c(\mathbf{r}, t), \quad (42)$$

which leads to the deterministic chemotactic term in the density equation.

The Brownian noise contribution requires an Itô expansion. For a single particle,

$$d\delta(\mathbf{r} - \mathbf{r}_i) = -\nabla \delta(\mathbf{r} - \mathbf{r}_i) \cdot d\mathbf{r}_i + D \nabla^2 \delta(\mathbf{r} - \mathbf{r}_i) dt. \quad (43)$$

The first term on the right-hand side is the stochastic contribution. Summing over all particles, it can be written as the divergence of a random current:

$$\mathbf{J}_{\text{noise}}(\mathbf{r}, t) = \sqrt{2D} \sum_{i=1}^N \delta(\mathbf{r} - \mathbf{r}_i) \boldsymbol{\xi}_i(t). \quad (44)$$

The second term gives the diffusive contribution after summation:

$$\sum_{i=1}^N D \nabla^2 \delta(\mathbf{r} - \mathbf{r}_i) = D \nabla^2 \rho(\mathbf{r}, t). \quad (45)$$

The random current has the correlation

$$\langle J_{\text{noise},a}(\mathbf{r}, t) J_{\text{noise},b}(\mathbf{r}', t') \rangle = 2D \rho(\mathbf{r}, t) \delta_{ab} \delta(\mathbf{r} - \mathbf{r}') \delta(t - t'). \quad (46)$$

It can therefore be represented by a spatio-temporal white noise field $\boldsymbol{\Lambda}(\mathbf{r}, t)$ with unit variance:

$$\mathbf{J}_{\text{noise}}(\mathbf{r}, t) = \sqrt{2D\rho(\mathbf{r}, t)} \boldsymbol{\Lambda}(\mathbf{r}, t), \quad (47)$$

where

$$\langle \Lambda_a(\mathbf{r}, t) \Lambda_b(\mathbf{r}', t') \rangle = \delta_{ab} \delta(\mathbf{r} - \mathbf{r}') \delta(t - t'). \quad (48)$$

Collecting all terms, we arrive at the Dean equation:

$$\boxed{\partial_t \rho = -\alpha \nabla \cdot (\rho \nabla c) + D \nabla^2 \rho - \nabla \cdot [\sqrt{2D\rho} \boldsymbol{\Lambda}]} \quad (49)$$

Under the mean-field approximation, the noise term vanishes on average since $\langle \boldsymbol{\Lambda} \rangle = 0$, giving the Keller-Segel type equation

$$\partial_t \langle \rho \rangle = -\alpha \nabla \cdot (\langle \rho \rangle \nabla c) + D \nabla^2 \langle \rho \rangle. \quad (50)$$

However, this noise term must be retained when calculating fluctuation quantities such as the structure factor $S(k)$.

2.2 Calculation of variance for Ornstein-Uhlenbeck process

For an Ornstein-Uhlenbeck process described by the equation

$$\partial_t x(t) = -\lambda x(t) + \eta(t), \quad (51)$$

where $\eta(t)$ is a Gaussian white noise with zero mean and correlation $\langle \eta(t) \eta(t') \rangle = 2\Gamma \delta(t - t')$, the formal solution is

$$X(t) = X(0) e^{-\lambda t} + \int_{-\infty}^t e^{-\lambda(t-s)} \eta(s) ds. \quad (52)$$

After a long time, the initial condition becomes irrelevant:

$$X(t) \approx \int_{-\infty}^t e^{-\lambda(t-s)} \eta(s) ds. \quad (53)$$

The variance of $X(t)$ can be calculated as:

$$\begin{aligned} \langle X^2(t) \rangle &= \int_{-\infty}^t \int_{-\infty}^t e^{-\lambda(2t-s-s')} \langle \eta(s) \eta(s') \rangle ds ds' \\ &= 2\Gamma \int_{-\infty}^t e^{-2\lambda(t-s)} ds \\ &= \frac{\Gamma}{\lambda}. \end{aligned} \quad (54)$$

2.3 Structure factor from the Lyapunov equation

The complete density covariance follows from the steady-state covariance equation of the two-dimensional linear Ornstein–Uhlenbeck system,

$$\frac{d}{dt} \begin{pmatrix} \delta \rho_{\mathbf{k}} \\ \delta c_{\mathbf{k}} \end{pmatrix} = A(k) \begin{pmatrix} \delta \rho_{\mathbf{k}} \\ \delta c_{\mathbf{k}} \end{pmatrix} + \begin{pmatrix} \eta_{\mathbf{k}} \\ 0 \end{pmatrix}, \quad (55)$$

where

$$A(k) = \begin{pmatrix} -Dk^2 & \alpha \rho_0 k^2 \\ -1 & -(D_c k^2 - \mu) \end{pmatrix}. \quad (56)$$

Introduce

$$a = Dk^2, \quad b = \alpha \rho_0 k^2, \quad q = D_c k^2 - \mu, \quad \Gamma = D \rho_0 k^2, \quad (57)$$

so that

$$A = \begin{pmatrix} -a & b \\ -1 & -q \end{pmatrix}, \quad Q = \begin{pmatrix} 2\Gamma & 0 \\ 0 & 0 \end{pmatrix}. \quad (58)$$

Here Q is the covariance matrix of the additive noise for the conjugate pair $\mathbf{k}$ and $-\mathbf{k}$.

Write the steady covariance matrix as

$$C = \begin{pmatrix} x & y \\ y & z \end{pmatrix}, \quad (59)$$

where

$$x = \langle \delta \rho_{\mathbf{k}} \delta \rho_{-\mathbf{k}} \rangle = C_{\rho\rho}(k) = \rho_0 S(k). \quad (60)$$

The Lyapunov equation

$$AC + CA^T + Q = 0 \quad (61)$$

produces

$$ax - by = \Gamma, \quad (62a)$$

$$x + (a + q)y - bz = 0, \quad (62b)$$

$$y = -qz. \quad (62c)$$

Using $z = -y/q$ in Eq. (62b) gives

$$y = -\frac{xq}{q(a+q) + b}. \quad (63)$$

Substitution into Eq. (62a) then yields

$$x = \Gamma \frac{q(a+q) + b}{(a+q)(aq+b)}. \quad (64)$$

Because $S(k) = x/\rho_0$, the normalized structure factor is

$$S(k) = \frac{\Gamma}{\rho_0} \frac{q(a+q) + b}{(a+q)(aq+b)}. \quad (65)$$

Substituting the original variables and cancelling the common factor k^2 gives

$$S(k) = D \frac{\alpha\rho_0 k^2 + (D_c k^2 - \mu) [(D + D_c)k^2 - \mu]}{[(D + D_c)k^2 - \mu] [D(D_c k^2 - \mu) + \alpha\rho_0]}. \quad (66)$$

The corresponding unnormalized density spectrum is simply

$$C_{\rho\rho}(k) = \rho_0 S(k). \quad (67)$$

Equation (66) immediately gives $S(k) = 1$ for $\alpha = 0$, confirming the Poisson normalization, and it approaches unity at large k . For fixed $\mu < 0$,

$$\lim_{k \rightarrow 0} S(k) = \frac{D}{D - \frac{\alpha\rho_0}{\mu}}. \quad (68)$$

For $\mu = 0$, the separate expression in Eq. (34) must be used when discussing the long-wavelength limit.

Equation (66) is valid only when both eigenvalues of $A(k)$ have negative real parts. The stability conditions are

$$a + q > 0, \quad aq + b > 0, \quad (69)$$

which are equivalent to

$$(D + D_c)k^2 - \mu > 0, \quad D(D_c k^2 - \mu) + \alpha\rho_0 > 0 \quad (70)$$

for nonzero k . If these conditions fail, the Lyapunov equation does not describe a stationary positive-definite covariance, and a steady-state structure factor is not defined within the linear theory.

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